Jacopo Di Silvestre

Disease Simulation Modeling and Cost-Effectiveness Analyses

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Professional Summary

Health economics researcher and PhD candidate who builds and works on microsimulation models in C++ and R to evaluate cancer screening strategies and their cost-effectiveness. Experience includes individual-based model development and calibration. Research spans lung and cervical cancer screening cost-effectiveness, smoking behavior trends, and evidence synthesis. Presented at various conferences such as Society for Medical Decision Making (SMDM) 2024 and 2026, Cancer intervention and surveillance modeling network (CISNET) 2024 annual meeting, and IASLC-WCLC 2026.

Technical Skills

Programming: C++, R, Bash

Computing and workflow: Git, Slurm, High-Performance Computing (HPC), Visual Studio

Analysis and reporting tools: Excel, LaTeX, Microsoft Office 365

Research Experience

PhD Candidate Mar 2025 - Present

University of Oslo, Oslo, Norway

- Conduct research within the Norwegian Cancer Society (Kreftforeningen)-funded LC-CONTROL project on lung cancer screening in Norway.
- Develop and apply individual-based disease simulation models to identify optimal lung cancer screening strategies in Norway, evaluating their clinical and cost-effectiveness.
- Model age- and birth cohort-specific smoking behaviors to inform population simulations (2026), while investigating opportunities to improve screening efficiency through biomarkers, AI-assisted risk prediction, and smoking cessation interventions (2027).

Research Scholar Nov 2024 - Present

Harvard Medical School - Massachusetts General Hospital
Center for Health Technology Assessment, Boston, MA

- Development and calibration of the Lung Cancer Policy Model (LCPM), a CISNET microsimulation model, to support analyses in the United States and Norway.
- Apply simulation modeling and C++ programming in lung cancer research.
- On-site research visits: Nov 2024 - Jan 2025; Aug - Sep 2025; Oct 2026.

Research Assistant Jul 2024 - Feb 2025

University of Oslo, Oslo, Norway

- Developed an individual-based lung cancer natural history model in R to evaluate the cost-effectiveness of risk-based low-dose CT screening in a high-risk Norwegian population.
- Analyzed outputs from the CISNET Harvard-CC model in R for a cost-effectiveness study of cervical cancer screening strategies in Norway.
- Conducted a systematic review of the health impact and cost-effectiveness of one-dose human papillomavirus (HPV) vaccination compared with alternative schedules (Mark Jit, Jane J Kim, Jean-François Laprise, Marc Brisson, Kiesha Prem, Emily A Burger).

Research Experience (continued)

Research Intern Sep 2023 - Feb 2024
ASCERTAIN European Project

- Conducted a scoping review of statistical methods for estimating comparative clinical effectiveness using evidence from single-arm or non-randomized trials (Eline Aas, Emily A Burger).

Education

PhD Studies Mar 2025 - Present
University of Oslo, Norway

MS in Health Economics and Health Management Sep 2022 - Jun 2024
Erasmus University Rotterdam

- European Master in Health Economics and Management (EU-HEM), with exchange studies at the University of Oslo and the University of Bologna.
- **Thesis:** "Risk-Based Lung Cancer Screening: A Microsimulation Model for the Norwegian High-Risk Population.", supervisors Emily A. Burger and Natalia Kunst
- Through an individual-based natural history model of lung cancer, this thesis analysis estimated NOK 129,689 per quality-adjusted life-year (QALY) gained for the evaluated risk-based screening strategy versus no screening in the modeled high-risk Norwegian population.
- **Relevant coursework:** Advanced Health Economic Modeling; Health Technology Assessment; Pharmaceutical Pricing and Market Access; Econometrics; Discrete Choice Experiments.

BS in Economics 2019 - 2022
University of Pavia, Italy

- **Thesis:** "Innovation and Competition Policy: U.S. Dynamism Analysis", supervisor prof. Patrizio Tirelli
- **Relevant coursework:** Econometrics, Microeconomics, Macroeconomics, and Monetary policy.

  Grade: 109/110.

Selected Publications

Di Silvestre J, Salgado MV, Jeon J, Mok Y, Holford TR, Vedøy TF, Pedersen K, Burger EA, Meza R. Age- and sex-specific smoking trends by birth cohort in Norway, 1978-2024: an analysis of cross-sectional survey data. *The Lancet Regional Health - Europe*. 2026;71:101886.
doi: 10.1016/j.lanepe.2026.101886

Pedersen K, **Di Silvestre J**, Sy S, Portnoy A, Castle PE, Kim JJ, Burger EA. Optimizing Cervical Cancer Screening by Age at Vaccination for Human Papillomavirus: Health and Resource Implications. *Annals of Internal Medicine*. 2026.
doi: 10.7326/ANNALS-25-03192

Teaching and Supervision

Supervise and mentor students in health economic modeling, providing methodological guidance, technical support, and critical feedback throughout their projects. Projects include developing cost-effectiveness models in R and Excel, conducting literature reviews, building system dynamics models, mapping clinical treatment pathways and estimating their downstream costs.